US 20250273176A1

(19) United States
(12) Patent Application Publication
(10) Pub. No.: US 2025/0273176 A1
(43) Pub. Date: Aug. 28, 2025
CHENG et al.

(54) **IMAGE PROCESSING DEVICE AND IMAGE DISPLAY METHOD**

(71) Applicant: **E Ink Holdings Inc.**, HSINCHU (TW)

(72) Inventors: **Hsiao-Lung CHENG**, HSINCHU (TW); **Shu-Cheng LIU**, HSINCHU (TW); **Pei-Lin TIEN**, HSINCHU (TW)

(21) Appl. No.: **19/051,196**

(22) Filed: **Feb. 12, 2025**

(30) **Foreign Application Priority Data**

Feb. 23, 2024 (TW) 113106685

Publication Classification

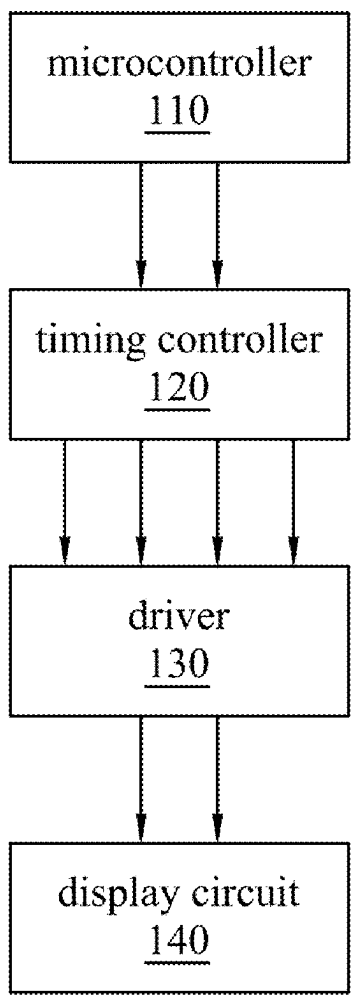
(51) **Int. Cl.**
G09G 3/34 (2006.01)

(52) **U.S. Cl.**
CPC ***G09G 3/344*** (2013.01); *G09G 2310/06* (2013.01); *G09G 2330/06* (2013.01)

(57) **ABSTRACT**

An image processing device, comprising a controller and a driver. The controller is configured to provide an image data and a plurality of waveform data. The image data comprises a plurality of pixel values of a plurality of frame periods, and the plurality of waveform data corresponds to the plurality of frame periods. The driver is coupled to the controller and a display circuit, and is configured to receive the image data and a first part of the plurality of waveform data from the controller. After the driver drives the display circuit according to the first part of the plurality of waveform data, the driver receives a second part of the waveform data from the controller.

100



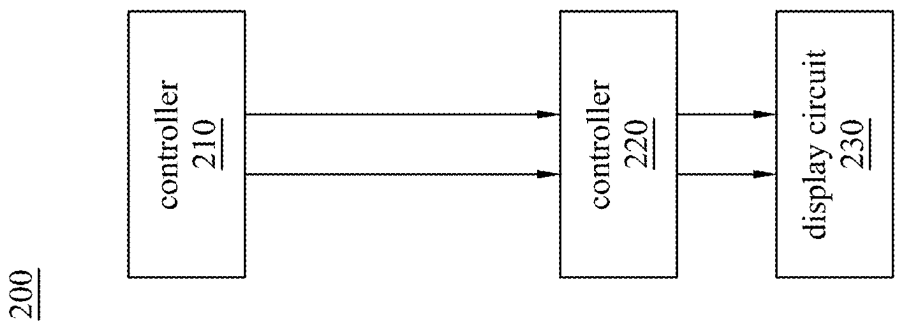

200
controller
210
controller
220
display circuit
230

Fig. 2

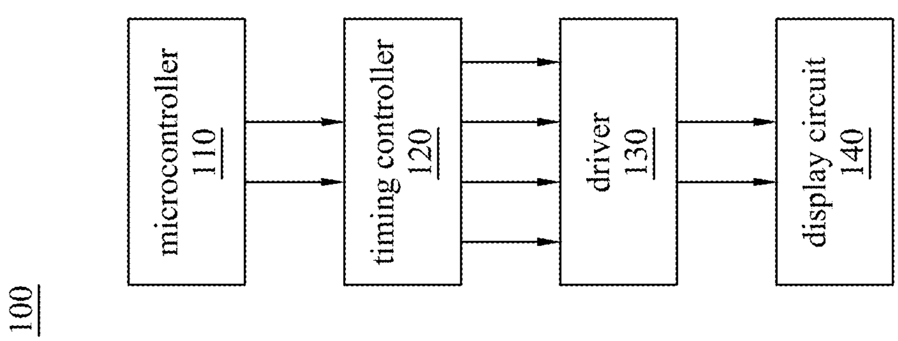

100
microcontroller
110
timing controller
120
driver
130
display circuit
140

Fig. 1

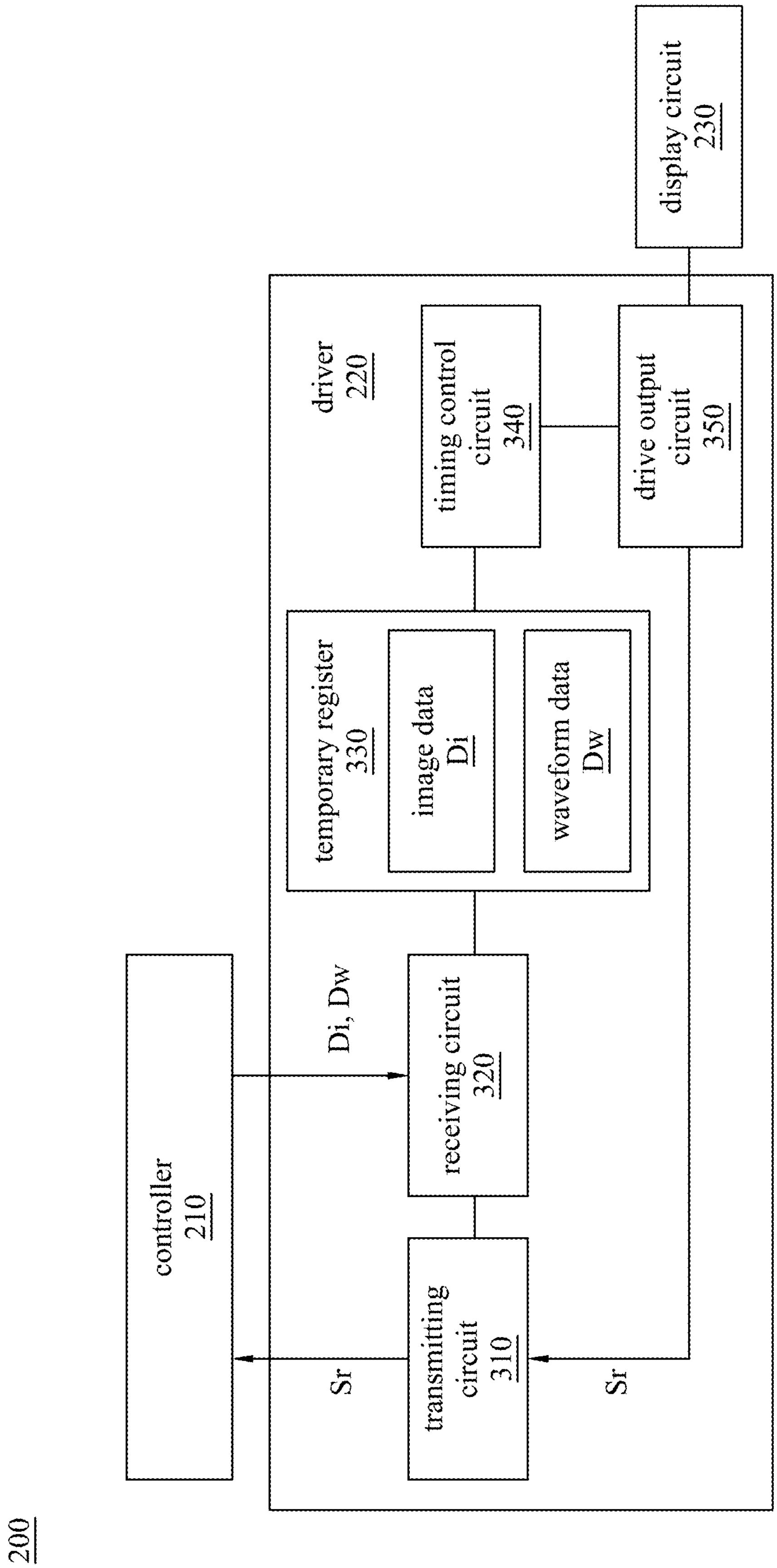

200
controller
210
driver
220
transmitting circuit
310
receiving circuit
320
temporary register
330
image data
Di
waveform data
Dw
timing control circuit
340
drive output circuit
350
display circuit
230
Di, Dw
Sr
Sr

Fig. 3

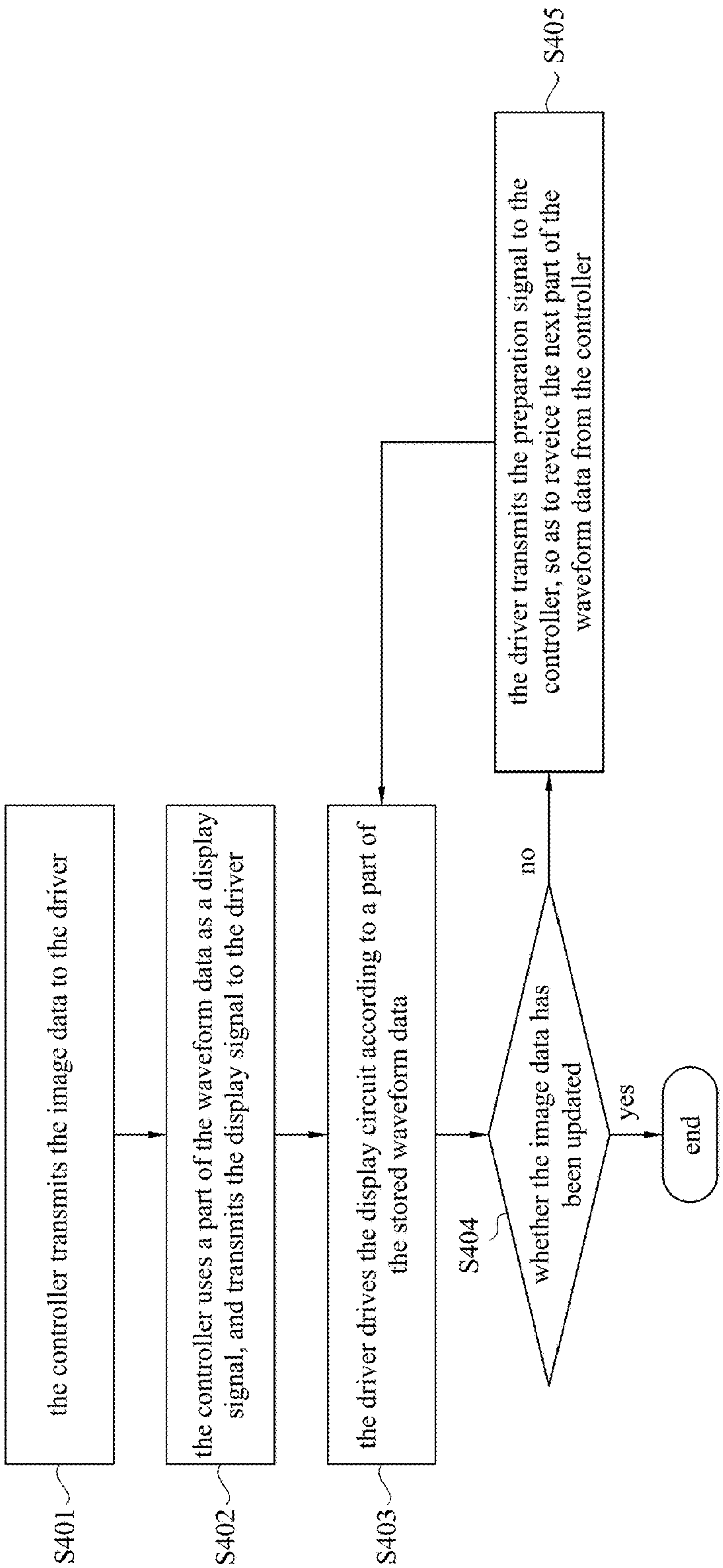

S401
the controller transmits the image data to the driver
S402
the controller uses a part of the waveform data as a display signal, and transmits the display signal to the driver
S403
the driver drives the display circuit according to a part of the stored waveform data
S404
whether the image data has been updated
yes
end
no
S405
the driver transmits the preparation signal to the controller, so as to reveice the next part of the waveform data from the controller

Fig. 4

IMAGE PROCESSING DEVICE AND IMAGE DISPLAY METHOD

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Taiwan Application Serial Number 113106685, filed Feb. 23, 2024, which is herein incorporated by reference in its entirety.

BACKGROUND

Technical Field

[0002] The present disclosure relates to image display technology, more particularly an image processing device and an image display method.

Description of Related Art

[0003] In various consumer electronics markets, “reflective display device” is widely used to implement the display screen, such as electronic paper display devices. The reflective display device mainly uses incident light to illuminate a display medium layer to achieve the purpose of display, so it can save power. However, as the image color or resolution increases, the size of the image data transmitted by the display device also increases, which may have a negative impact within the display device.

SUMMARY

[0004] One aspect of the present disclosure is an image processing device, comprising a controller and a driver. The controller is configured to provide an image data and a plurality of waveform data. The image data comprises a plurality of pixel values of a plurality of frame periods, and the plurality of waveform data corresponds to the plurality of frame periods. The driver is coupled to the controller and a display circuit, and is configured to receive the image data and a first part of the plurality of waveform data from the controller. After the driver drives the display circuit according to the first part of the plurality of waveform data, the driver receives a second part of the waveform data from the controller.

[0005] Another aspect of the present disclosure is an image display method, comprising: transmitting, by a controller, an image data to a driver, wherein the image data comprises a plurality of pixel values of a plurality of frame periods; transmitting, by the controller, a first part of a plurality of waveform data to the driver, wherein the plurality of waveform data corresponds to the plurality of frame periods; driving, by the driver, a display circuit according to the first part of the plurality of waveform data; and receiving, by the driver, a second part of the waveform data from the controller.

[0006] Another aspect of the present disclosure is an image processing device, comprising a controller and a driver. The controller is configured to provide an image data and a plurality of waveform data. The image data comprises a plurality of pixel values of a plurality of frame periods, and the plurality of waveform data corresponds to the plurality of frame periods. The driver is coupled to the controller and a display circuit, and is configured to receive the image data and a first part of the plurality of waveform data from the controller. When the driver encodes the first part of the plurality of waveform data to generate a plurality of voltage data, the driver receives a second part of the waveform data from the controller.

[0007] It is to be understood that both the foregoing general description and the following detailed description are by examples, and are intended to provide further explanation of the disclosure as claimed.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The present disclosure can be more fully understood by reading the following detailed description of the embodiment, with reference made to the accompanying drawings as follows:

[0009] FIG. **1** is a schematic diagram of an image processing device in some embodiments of the present disclosure.

[0010] FIG. **2** is a schematic diagram of an image processing device in some embodiments of the present disclosure.

[0011] FIG. **3** is a schematic diagram of a driver in some embodiments of the present disclosure.

[0012] FIG. **4** is a flowchart illustrating an image display method in some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0013] For the embodiment below is described in detail with the accompanying drawings, embodiments are not provided to limit the scope of the present disclosure. Moreover, the operation of the described structure is not for limiting the order of implementation. Any device with equivalent functions that is produced from a structure formed by a recombination of elements is all covered by the scope of the present disclosure. Drawings are for the purpose of illustration only, and not plotted in accordance with the original size.

[0014] It will be understood that when an element is referred to as being “connected to” or “coupled to”, it can be directly connected or coupled to the other element or intervening elements may be present. In contrast, when an element to another element is referred to as being “directly connected” or “directly coupled,” there are no intervening elements present. As used herein, the term “and/or” includes an associated listed items or any and all combinations of more.

[0015] FIG. **1** is a schematic diagram of an image processing device **100** in some embodiments of the present disclosure, which is used to implement a display device (e.g., reflective display device, electronic paper display). The image processing device **100** includes a microcontroller **110**, a timing controller **120**, a driver **130** and a display circuit **140** (e.g., display panel). The microcontroller **110** is configured to provide an image data. The timing controller **120** stores a waveform look up table, and the waveform look up table records multiple voltage waveforms corresponding to different pixel values. The timing controller **120** is configured to obtain the corresponding voltage waveform according to the image data, and apply voltage to multiple pixcel circuits in the display circuit **140** through the driver **130**, so as to display the expected the pixel value.

[0016] The above “image data” can be an image screen that needs to be updated by the display circuit **140**. For example, the image data includes multiple pixel values of multiple frame periods displayed by the display circuit **140**.

In order for each pixel circuit of the display circuit **140** to display the expected pixel value, the driver **130** needs to provide different voltages sequentially in multiple frame periods. For example, in order to display the pixel value "155", the driver **130** needs to provide voltages of 3V, 0V, and 2V sequentially in three frame periods. The waveform formed by multiple voltages is called "waveform data" here.

[0017] For example, each pixel value is controlled by a 2-bit waveform. Under the condition that the resolution is 2560×1600, the refresh rate is 75 Hz, and the data bus in the image processing device **100** is 16 bits, the signal transmission frequency will be 38.4 MHZ (2560×1600×75×2/16), which may cause electromagnetic interference (EMI) problems.

[0018] FIG. **2** is a schematic diagram of an image processing device **200** in some embodiments of the present disclosure. The image processing device **200** includes a controller **210** and a driver **220**, and is coupled to a display circuit **230**. In one embodiment, the image processing device **200** is implemented to a display device, and the display circuit **230** can be a reflective display panel, such as an electrophoretic display device (electronic paper), but the present disclosure is not limited to this. The electrophoretic display device includes a transistor array layer and an electronic ink layer. The transistor array layer (e.g., thin film transistor array, TFT array) can form an electric field according to a control voltage to adjust multiple positions of multiple electrophoretic particles in the electronic ink layer, so as to present different gray scales or different colors. The electronic ink layer includes multiple electrophoretic particles, and the electrophoretic particles belong to different types (e.g., black, white), which are separately encapsulated in multiple microcapsules or microcups to form a pixel unit.

[0019] The controller **210** is configured to provide the image data, and stores the complete waveform data. As mentioned above, the image data includes multiple pixel values of multiple frame periods, and corresponds to an image screen that needs to be updated by the display circuit **230**. Each of waveform data corresponds to each frame period, and is configured to represent the driving voltage that needs to be provided to the display circuit **230** in each frame.

[0020] The driver **220** is coupled to the controller **210** and the display circuit **230**, and is configured to receive the image data and the waveform data from the controller **210** to drive the display circuit **230**. In order to avoid electromagnetic interference caused by high signal transmission frequency, the controller **210** transmits the waveform data to the driver **220** in batches. In other words, the controller **210** stores complete waveform data, but the driver **220** only needs to store a part of the waveform data. Accordingly, the frequency of signal transmission can be reduced.

[0021] The driver **220** is configured to apply voltage to the display circuit **230** according to the waveform data, and each part of the waveform data corresponds to one or more frames. Specifically, the controller **210** first transmits a first part of the waveform data to the driver **220**, when the driver **220** starts to decode the first part of the waveform data, or drives the display circuit **230** according to the first part of the waveform data, the controller **210** transmits a second part of the waveform data to the driver **220**. Each part of the waveform data can be voltage data (e.g., driving voltage) of one or more frames. For example, the controller **210** first transmits a waveform of a first frame (first part) to the driver **220**, then when the driver **220** controls the display circuit **230** to perform the update during the first frame, the controller **210** transmits a waveform of a second frame (second part) to the driver **220**.

[0022] FIG. **3** is a schematic diagram of a driver **220** in some embodiments of the present disclosure. In one embodiment, the driver **220** includes a transmitting circuit **310**, a receiving circuit **320**, a temporary register **330**, a timing control circuit **340** and a drive output circuit **350**. The transmitting circuit **310** and the receiving circuit **320** are respectively coupled to the controller **210**, and are configured to transfer data. The temporary register **330** may be a buffer register configured to store the image data Di and parts of the waveform data Dw received from the controller **210**. The timing control circuit **340** is configured to decode the waveform data Dw to generate the corresponding voltage data (e.g., driving voltage), and provide the voltage data to the display circuit **230** through the drive output circuit **350**, so as to control the display circuit **230** to present/display the corresponding pixel value.

[0023] In one embodiment, after the drive output circuit **350** drives display circuit **230** according to a part of the waveform data Dw (i.e., after providing the voltage data to the display circuit **230**), the drive output circuit **350** transmits a preparation signal Sr to the controller **210** through the transmitting circuit **310**, so as to notify the controller **210** to transmit the next part of the waveform data. In other embodiments, when each of frame periods ends, the drive output circuit **350** transmits the preparation signal Sr to the controller **210** through the transmitting circuit **310**, so as to notify the controller **210** can transmit the next part of the waveform data Dw.

[0024] In some embodiments, the controller **210** can transmit the next part of the waveform data Dw to the driver **220** when the driver **220** starts to decode a part of the waveform data Dw to generate the voltage data without waiting for the end of the current frame. In other words, the controller **210** can transmit each part of the waveform data Dw in batches according to a preset time interval, and is not limited to transmitting the waveform data Dw in response to the preparation signal Sr. For example, after the timing control circuit **340** decodes the waveform data Dw of the first frame (first part), the receiving circuit **320** can receive the waveform data Dw of the second frame (second part) from the controller **210**. Then, when the drive output circuit **350** controls the display circuit **230** to complete the update of the first frame and transmits the preparation signal Sr through the transmitting circuit **310**, the controller **210** transmits the waveform data Dw of the third frame (third part) to the receiving circuit **320**.

[0025] FIG. **4** is a flowchart illustrating an image display method in some embodiments of the present disclosure. Referring to FIG. **2** and FIG. **3**, in step S**401**, the controller **210** first transmits the image data Di to the driver **220**. As mentioned above, the image data Di corresponds to an image screen that needs to be updated during multiple frames by the display circuit **230**, so the image data Di includes multiple pixel values of multiple frame periods.

[0026] In step S**402**, the controller **210** uses a part of the waveform data Dw as a display signal, and transmits the display signal to the driver **220**. Each waveform data Dw corresponds to each frame period, and every part of the waveform data Dw transmitted by the controller **210** is the waveform that needs to use in one or more frame periods.

[0027] In step S**403**, the driver **220** generates a driving voltage according to a part of the image data Di and the waveform data Dw stored in the temporary register **330** to drive the display circuit **230**. In one embodiment, the display circuit **230** is an electrophoretic display device, and is configured to form an electric field according to the received voltage data (e.g., driving voltage) to adjust positions of multiple electrophoretic particles in the electronic ink layer.

[0028] In step S**404**, the driver **220** determines whether the image data Di has been completely updated (e.g., determine whether the waveform data of all the frame periods has been received). If the update has been completed, end the current display process.

[0029] If the driver **220** determines the image data Di has not been completely updated, in step S**405**, after the driver **220** drives the display circuit **230** through the drive output circuit **350**, the drive output circuit **350** transmits the preparation signal Sr to the controller **210** through the transmitting circuit **310** at the same time, so as to receive the next part of the waveform data Dw from the controller **210**.

[0030] In the aforementioned embodiment, the controller **210** transmits each part of the waveform data Dw to the driver **220** according to the preparation signal Sr returned by the driver **220**, but the present disclosure is not limited to this. In some embodiments, the controller **210** may also actively transmit each part of the waveform data Dw to the driver **220** according to a preset time interval. For example, the controller **210** first transmits the waveform data Dw of the first frame to the driver **220**, then after the preset time interval, when the driver **220** decodes the waveform data Dw of the first frame, the controller **210** transmits the waveform data Dw of the second frame to the driver **220**, so that the driver **220** stores the waveform data Dw of the second frame in advance.

[0031] As mentioned above, after the driver **220** drives the display circuit **230** according to the waveform data Dw of the first frame, the driver **220** transmits the preparation signal Sr to the controller **210** to receive the waveform data Dw of the third frame from the controller **210**. At this time, since the driver **220** has received the waveform data Dw of the second frame, the driver **220** can start to decode the waveform data Dw of the second frame while receiving the waveform data Dw of the third frame, so as to drive the display circuit **230**.

[0032] The present disclosure stores the complete waveform data Dw in the controller **210**, and transmits each part of the waveform data Dw to the driver **220** in batches and sequentially. In other words, only a part of the waveform data Dw is transmitted between the controller **210** and the driver **220** at a time, and the driver **220** only needs to store the part currently to be processed and the part to be processed soon of the waveform data Dw. Therefore, the signal transmission frequency in the image processing device **200** will be controlled to avoid possible electromagnetic interference.

[0033] The elements, method steps, or technical features in the foregoing embodiments may be combined with each other, and are not limited to the order of the specification description or the order of the drawings in the present disclosure.

[0034] It will be apparent to those skilled in the art that various modifications and variations can be made to the structure of the present disclosure without departing from the scope or spirit of the present disclosure. In view of the foregoing, it is intended that the present disclosure cover modifications and variations of this present disclosure provided they fall within the scope of the following claims.

What is claimed is:

1. An image processing device, comprising:

a controller configured to provide an image data and a plurality of waveform data, wherein the image data comprises a plurality of pixel values of a plurality of frame periods, and the plurality of waveform data corresponds to the plurality of frame periods; and

a driver coupled to the controller and a display circuit, and configured to receive the image data and a first part of the plurality of waveform data from the controller, wherein after the driver drives the display circuit according to the first part of the plurality of waveform data, the driver receives a second part of the waveform data from the controller.

2. The image processing device of claim **1**, wherein after the driver drives the display circuit according to the first part of the plurality of waveform data, the driver transmits a preparation signal to the controller to obtain the second part of the waveform data from the controller.

3. The image processing device of claim **1**, wherein when each of the plurality of frame periods ends, the driver transmits a preparation signal to the controller to obtain the second part of the waveform data from the controller.

4. The image processing device of claim **1**, wherein the driver is configured to apply voltage to the display circuit according to the plurality of waveform data, and the first part of the plurality of waveform data corresponds to one or more of the plurality of frame periods.

5. The image processing device of claim **1**, wherein the driver is configured to decode the plurality of waveform data to control the display circuit to display the plurality of pixel values.

6. An image display method, comprising:

transmitting, by a controller, an image data to a driver, wherein the image data comprises a plurality of pixel values of a plurality of frame periods;

transmitting, by the controller, a first part of a plurality of waveform data to the driver, wherein the plurality of waveform data corresponds to the plurality of frame periods;

driving, by the driver, a display circuit according to the first part of the plurality of waveform data; and

receiving, by the driver, a second part of the waveform data from the controller.

7. The image display method of claim **6**, further comprising:

transmitting, by the driver, a preparation signal to the controller to obtain the second part of the waveform data.

8. The image display method of claim **6**, wherein receiving the second part of the waveform data from the controller comprises:

when each of the plurality of frame periods ends, transmitting a preparation signal to the controller to obtain the second part of the waveform data.

9. The image display method of claim **6**, wherein the driver is configured to apply voltage to the display circuit according to the plurality of waveform data, and the first part of the plurality of waveform data corresponds to one or more of the plurality of frame periods.

10. The image display method of claim **6**, wherein the driver is configured to decode the plurality of waveform data to control the display circuit to display the plurality of pixel values.

11. An image processing device, comprising:

a controller configured to provide an image data and a plurality of waveform data, wherein the image data comprises a plurality of pixel values of a plurality of frame periods, and the plurality of waveform data corresponds to the plurality of frame periods; and

a driver coupled to the controller and a display circuit, and configured to receive the image data and a first part of the plurality of waveform data from the controller, wherein when the driver encodes the first part of the plurality of waveform data to generate a plurality of voltage data, the driver receives a second part of the waveform data from the controller.

12. The image processing device of claim **11**, wherein after the driver drives the display circuit according to the first part of the plurality of waveform data, the driver transmits a preparation signal to the controller to obtain a third part of the waveform data from the controller.

13. The image processing device of claim **11**, wherein the driver is configured to apply voltage to the display circuit according to the plurality of waveform data, and the first part of the plurality of waveform data corresponds to one or more of the plurality of frame periods.

14. The image processing device of claim **11**, wherein the display circuit is an electrophoretic display device, and is configured to adjust a plurality of positions of a plurality of electrophoretic particles in an electronic ink layer according to the plurality of voltage data.

15. The image processing device of claim **14**, wherein the electronic ink layer includes a plurality of electrophoretic particles, and the plurality of electrophoretic particles belongs to different types.

* * * * *